

VPC Part 2

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Topics

1. IP addressing

- VPC CIDR blocks
- Subnet CIDR blocks

1. Virtual private clouds

- Default VPCs
- Create a My VPC
- Configure your VPC

1. Subnets

4. Connect VPC to other networks

- Internet gateways
- Egress-only internet gateways
- NAT devices
- Elastic IP addresses
- AWS Transit Gateway
- AWS Virtual Private Network
- VPC peering connections

4. Monitoring

5. Monitoring

6. Security

7. Use with other services

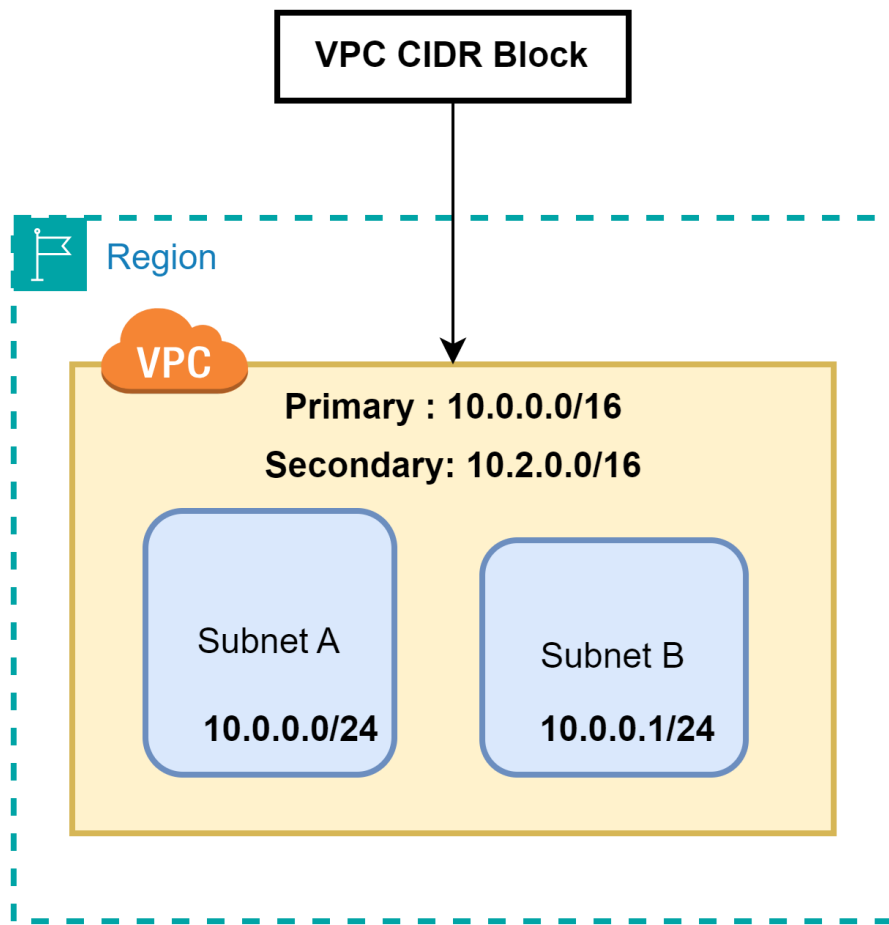
IP addressing

VPC CIDR blocks

- Allowed block size /16 netmask (65,536 IP addresses) to /28 netmask (16 IP addresses).
- Associate CIDR to VPC.
- Optionally attach additional VPC
- You can create publically routable CIDR block

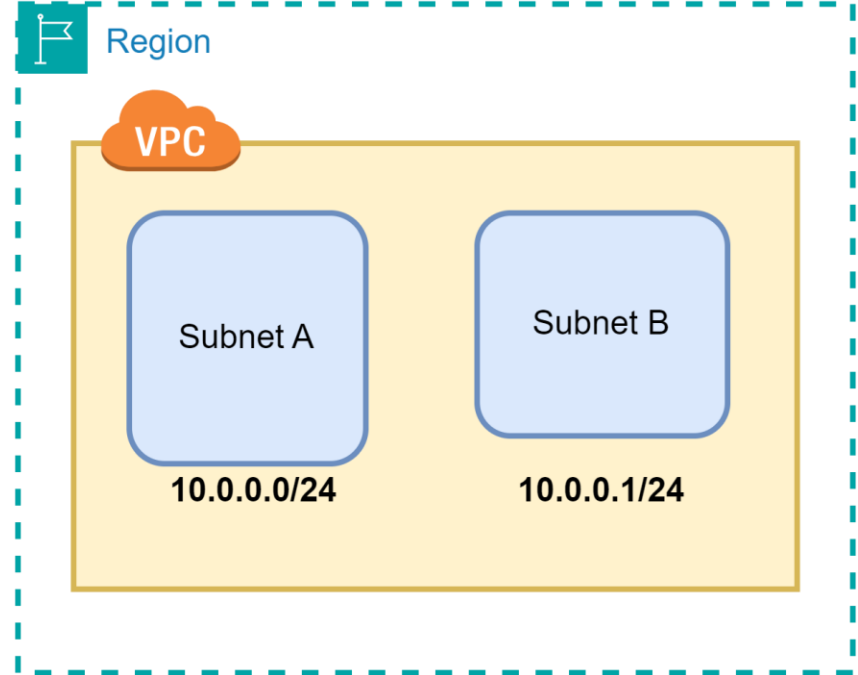
Local Routes for VPC:

Destination	Target
10.0.0.0/16	Local
10.2.0.0/16	Local



Subnet CIDR blocks

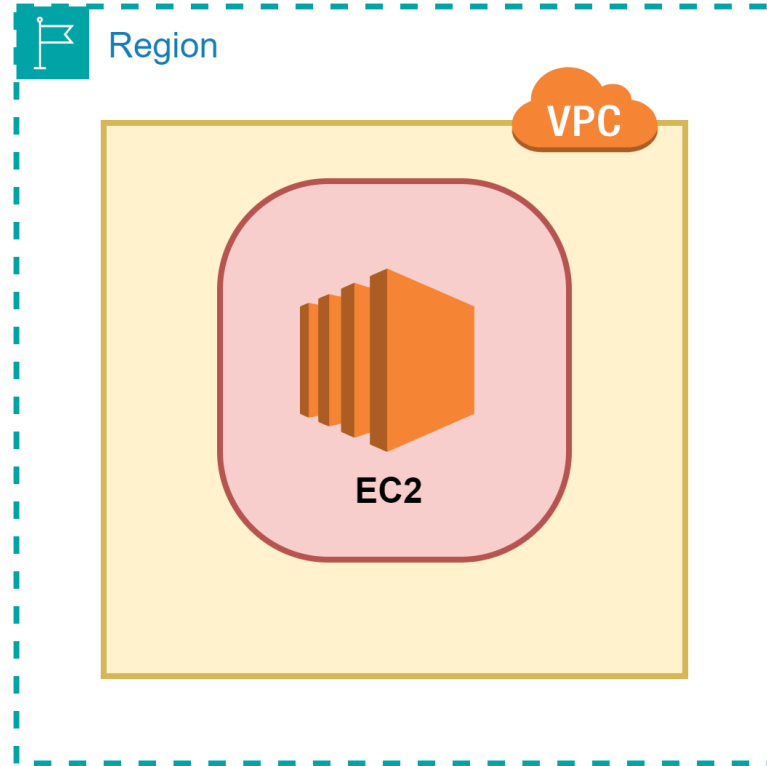
- The IP addresses represented using Classless Inter-Domain Routing (CIDR) notation.
- CIDR block of subnet can be same as VPC CIDR.
- You can create and attach multiple subnet to VPC
- If you create more than one subnet in a VPC, the CIDR blocks of the subnets cannot overlap.
- CIDR block size for a subnet is between a /28 netmask and /16 netmask
- First four IP addresses and the last IP address in each subnet CIDR block are not available for your use



Virtual private clouds

Default AWS VPCs.

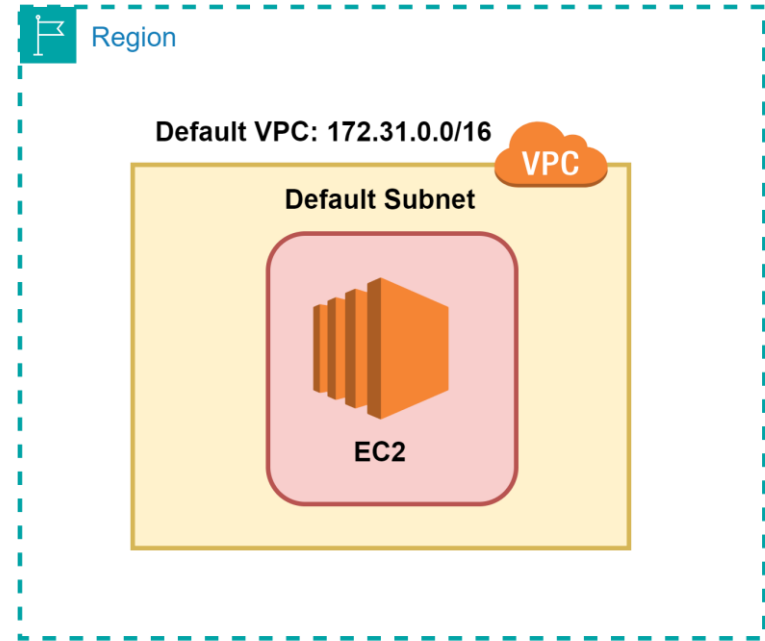
- Default VPC in each aws **Region**
- It is suitable for getting started **quickly** and for launching public instances such as a blog or simple website.
- Services can use like EC2, Elastic Load Balancing, Amazon RDS, and Amazon EMR
- It comes with a public subnet in each Availability Zone, an internet gateway, and settings to enable DNS resolution.
- You can add subnets to your default VPC



Exercise of default VPC

Steps:

1. Login to the Console - EC2
2. Launch Instances - Enter instance Name
3. Select the Amazon Linux AMI(Image)
4. Select the Instance Type(Free tier eligible if needed)
5. Create new key pair/select existing key pair
6. Network Settings - Select default vpc
7. No preference (Default subnet in any availability zone)
8. Select the default/existing security group under Common security groups
9. Click it on Launch Template
10. SSH EC2 Instance with .pem key



Create a My VPC

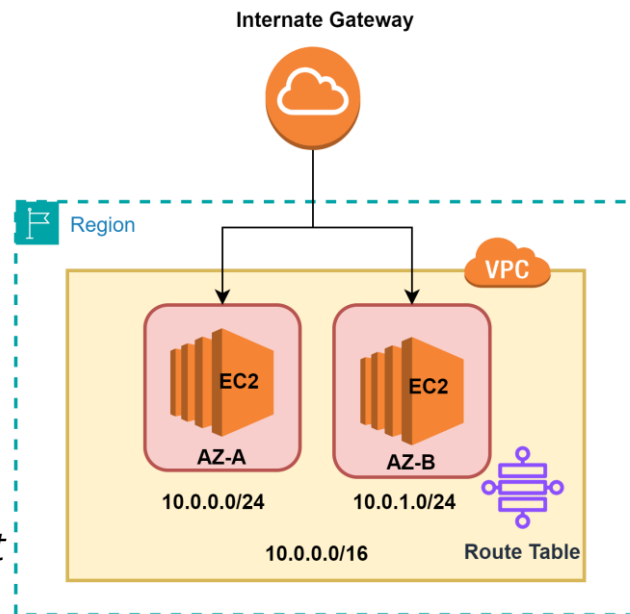
Contents :

VPC configuration options :

- Availability Zones (multiple AZs): Discrete data centers
 - CIDR blocks: Specify IP address ranges
 - DNS options: Enable DNS hostnames(public ipv4), Enable DNS resolution(Route53)
- Internet gateway: Connects your VPC to the internet
- NAT gateways: Access internet from private subnet instance
- Route tables: Route the traffic
- Subnet
- Tenancy: Allocate dedicated hardware for your vpc

Create a VPC using the AWS CLI

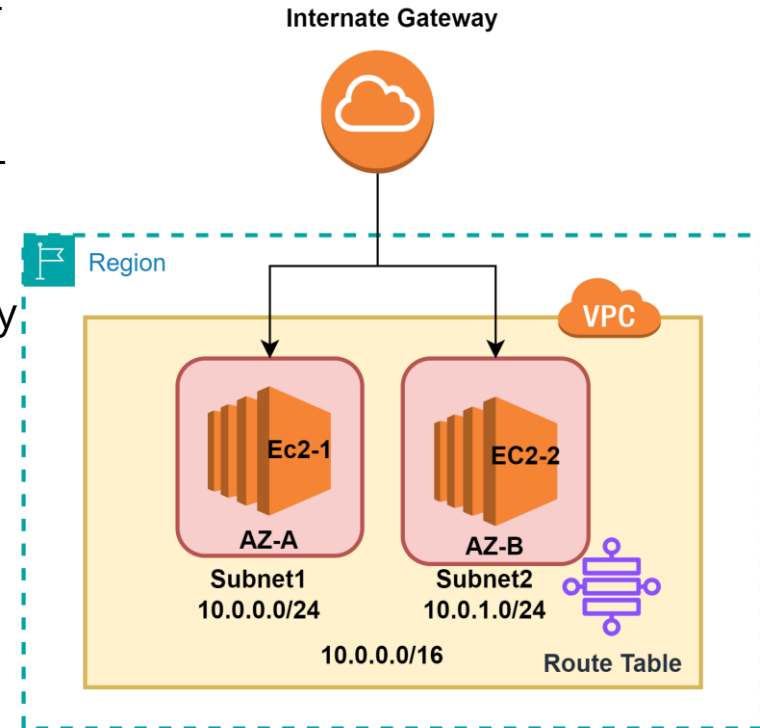
```
aws ec2 create-vpc --cidr-block 10.0.0.0/24 --query Vpc.VpcId --output text
```



Exercise with my VPC

Steps:

1. **My VPC** : Enter VPC Name & CIDR block
2. **Subnet1** : Select my vpc id -Enter Subnet name - Select Availability Zone A like us-east-1a -Select VPC CIDR block - Enter subnet CIDR block(10.0.0.0/24)
3. **Subnet2** : Select my vpc id -Enter Subnet name - Select Availability Zone B like us-east-1b -Select VPC CIDR block - Enter subnet CIDR block(10.0.1.0/24)
4. **Ec2-1** : Enter Name-Select AMI,instance type, vpc key pair-Select my vpc, subnet1 and Enter the security group name and inbound rule(SSH,HTTP,HTTPS)
5. **Ec2-2** : Enter the ec2 name-Select AMI,instance type, vpc key pair-Select my vpc, subnet2 and select existing security group.
6. **Route Table** : Select the my vpc
7. **Internet gateway** and attach to VPC (Allows communication between VPC and **internet**)



VPC Connectivity Options

- Internet gateways
- NAT devices
 - Allow resources in private subnets to connect to the internet, other VPCs, on-premises
 - a. NAT Gateway: offered by AWS
 - b. NAT instance: Use it on an Instance
- Elastic IP addresses
- AWS Transit Gateway
- AWS Virtual Private Network
- VPC peering connections
- Egress-only internet gateways
 - allows outbound communication
 - prevents the internet from initiating an IPv6 connection with your instances.

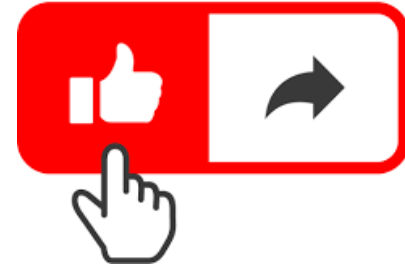
What is AWS VPC (Virtual Private Cloud)

VPC Component

Part2

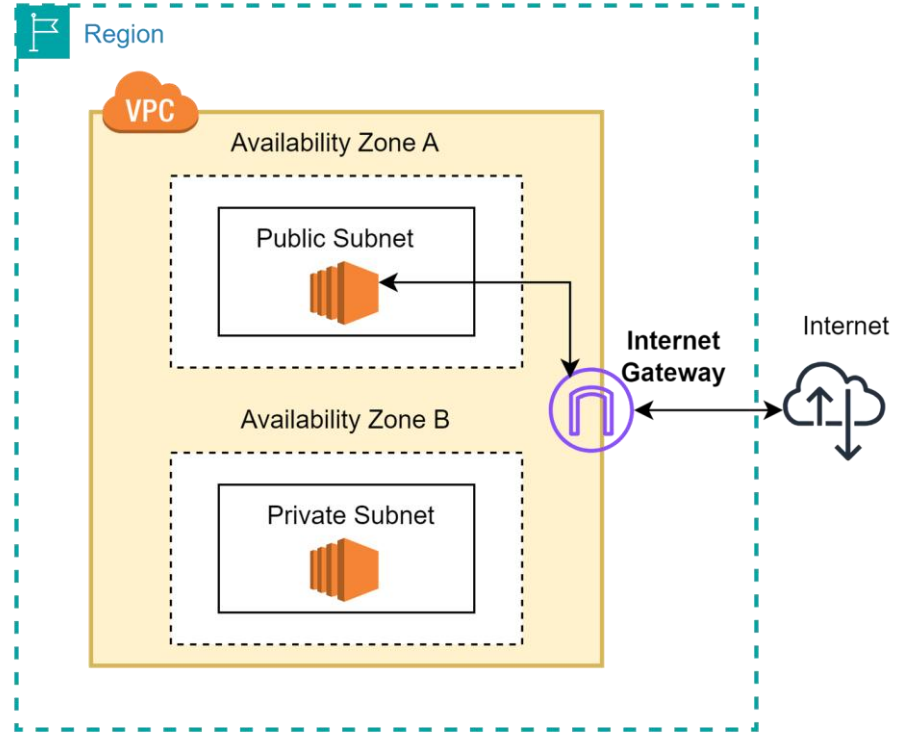


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Internet Gateway

- Allows communication between VPC and **internet**
 - Enables resources in public subnet
 - Provides a target in VPC route tables for internet-routable traffic
 - Supports IPv4 & IPv6 traffic
- Examples, an IGW enables you to connect to an EC2 instance in AWS using your local computer.

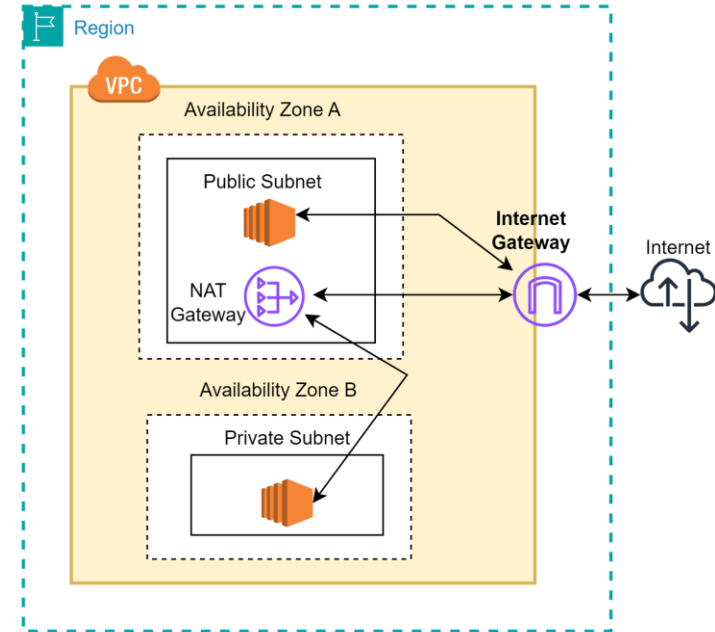


NAT Devices

Allow resources in private subnets to connect to the internet, other VPCs, on-premises

- a. NAT Gateway: offered by AWS
- b. NAT instance: Use it on an Instance

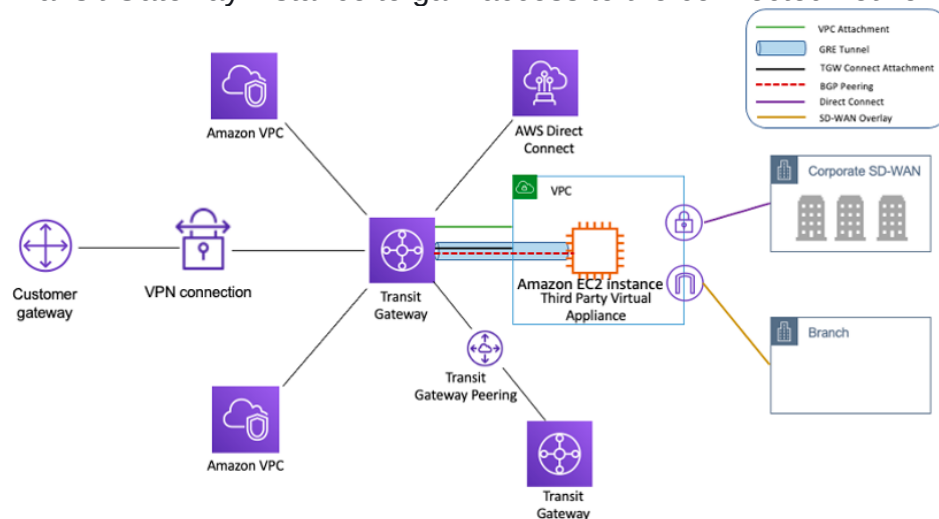
Note: Recommendation to Use NAT Gateway for better availability and bandwidth and require less effort on your part to administer.



AWS Transit Gateway

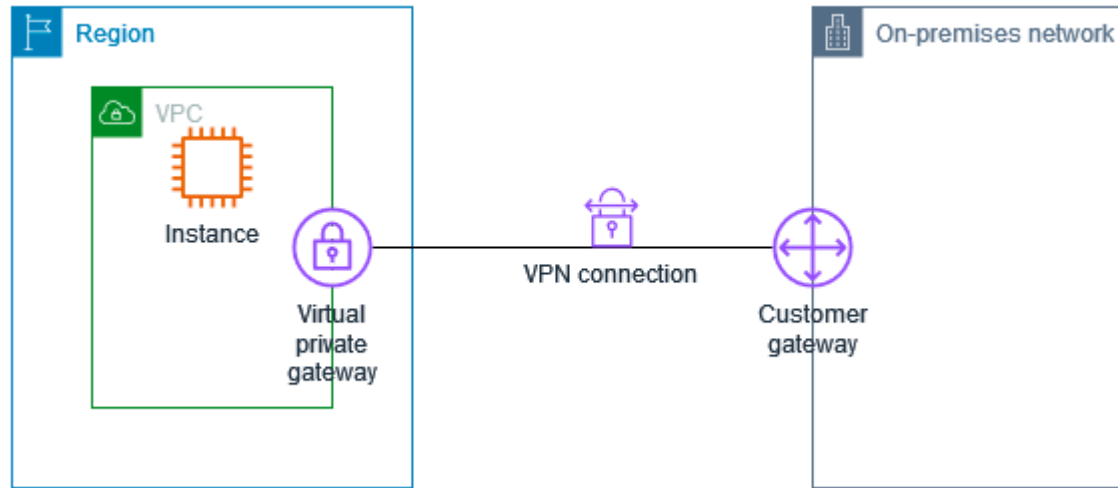
AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This connection simplifies your network and puts an end to complex peering relationships. Transit Gateway acts as a highly scalable cloud router—each new connection is made only once.

- It enables customers to connect thousands of VPCs.
- Can attach all your hybrid connectivity (VPN and Direct Connect connections) to a single Transit Gateway instance, consolidating and controlling your organization's entire AWS routing configuration in one place
- This hub-and-spoke [model](#) simplifies management and reduces operational costs because VPCs only connect to the Transit Gateway instance to gain access to the connected networks.



AWS Virtual Private Network

AWS Virtual Private Network (AWS VPN) establishes a secure and private tunnel from your network or device to the AWS Cloud. You can extend your existing on-premises network into a VPC, or connect to other AWS resources from a client. AWS VPN offers two types of private connectivity that feature the high availability and robust security necessary for your data.

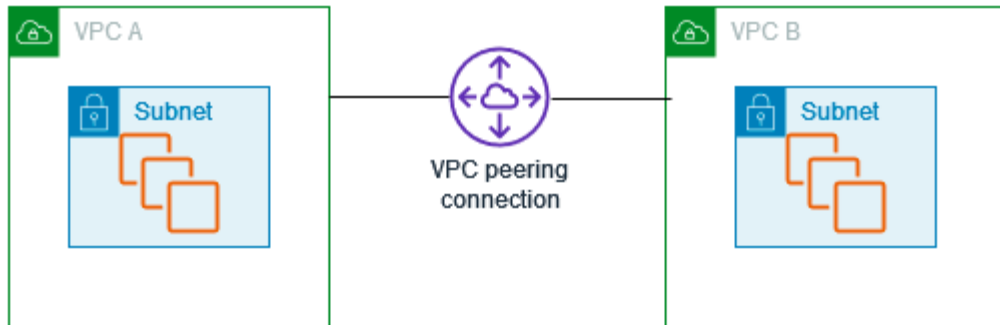


A virtual private gateway is the VPN concentrator on the Amazon side of the Site-to-Site VPN connection.

VPC peering connections

It is networking connection between **two VPCs** that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses

Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account.

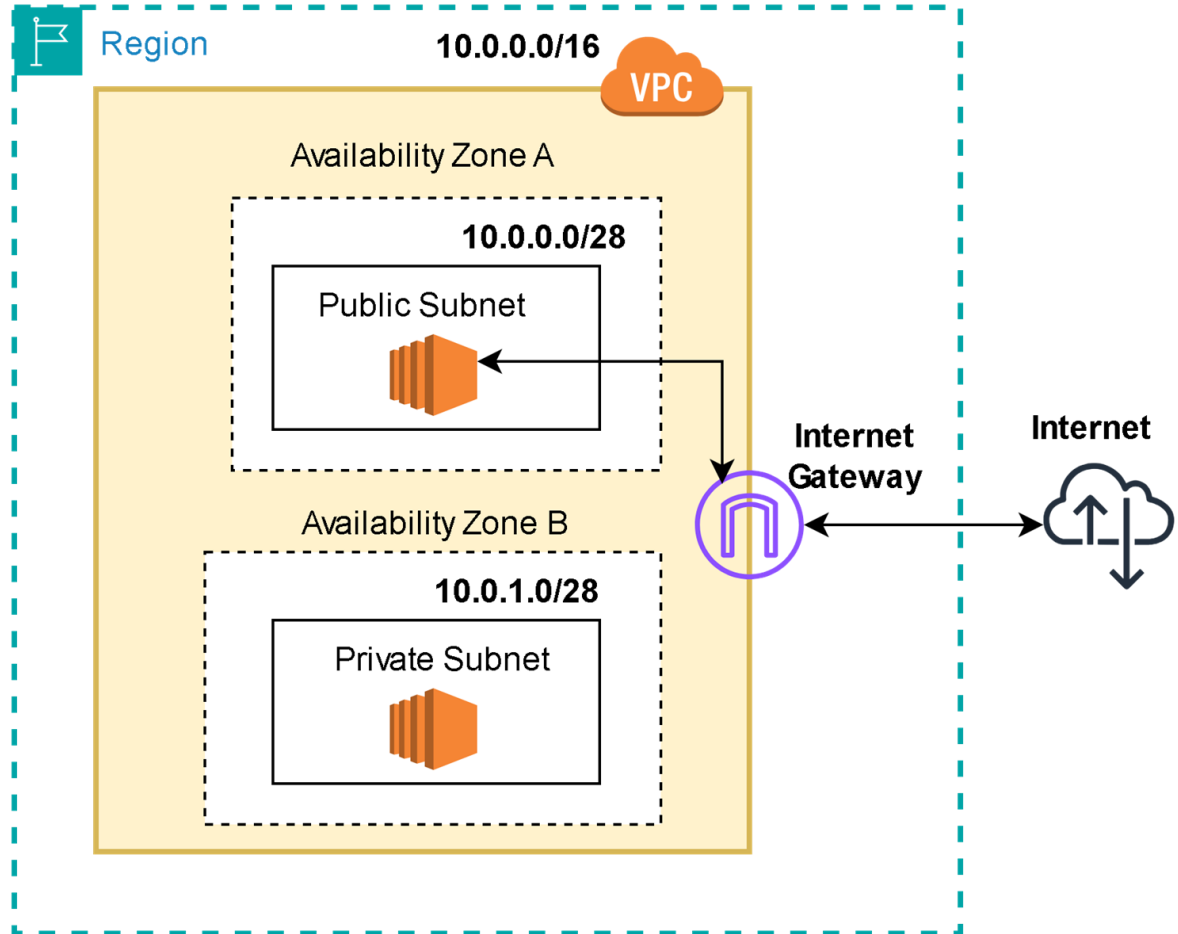


Elastic IP addresses

An Elastic IP address is a **public** static IPv4 address, which is reachable from the internet. If your instance does not have a public IPv4 address, you can associate an Elastic IP address with your instance to enable communication with the internet

Exercise:

1. Login AWS Console
2. Create VPC with 10.0.0.0/16 CIDR range
3. Create two subnets
4. Public Subnet association with Route Table
5. Create Security Group Rules for public subnet(0.0.0.0/0)
6. Create internet gateway and attach it to VPC
7. Create Route table and route the traffic to the internet
8. Create EC2 instances in private and public subnet



Exercise:

1. Create a VPC:

- Click on "Create VPC".
- Provide a name for your VPC (e.g., "My VPC").
- Enter the desired CIDR block for your VPC (e.g., 10.0.0.0/16). This will be the overall range for all your subnets within the VPC.
- Click "Create" to provision your VPC.

2. Create Public and Private Subnets:

- Create a public subnet with a name (e.g., "PublicSubnet") and a CIDR block within the VPC's range (e.g., 10.0.1.0/24). Choose an Availability Zone.
- Repeat the above process to create a private subnet with a name (e.g., "PrivateSubnet") and a separate CIDR block (e.g., 10.0.2.0/24) in a different Availability Zone.

3. Create an Internet Gateway (IGW):

- Enter a name for your IGW (e.g., "MyInternetGateway").
- Click "Create internet gateway" to provision it.

4. Attach the IGW to the VPC:

5. Create a NAT Gateway:

6. Create a Route Table for the Public Subnet:

7. Associate the Public Subnet Route Table with the Public Subnet:

8. Create a Route in the Public Subnet Route Table (Optional):

- While the route table might be automatically populated depending on your configuration, you can also manually create a route for internet access.
- Select the route table for the public subnet (e.g., "PublicRouteTable").
- Click on the "Routes" tab.
- Click on "Edit routes".
- Enter `0.0.0.0/0` in the "Destination" field (this represents all traffic).
- Select "Internet Gateway" in the "Target" field.
- Choose the internet gateway you created (e.g., "MyInternetGateway") from the dropdown menu.
- Click "Save".

9. Create a Private Route Table:

10. Associate the Private Subnet Route Table with the Private Subnet:

11. Create a Route in the Private Subnet Route Table:

- Select the route table for the private subnet (e.g., "PrivateRouteTable").

Thank You



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